

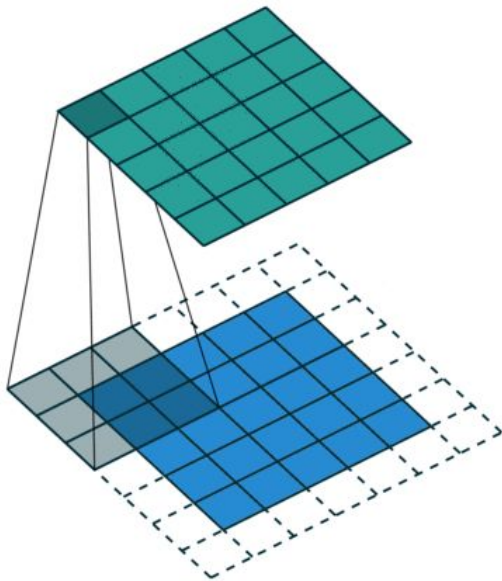
Multiscale and Hybrid Vision Architectures

Samsung Research Institute, Bangalore

8th June, 2026

CNNs:

Date



Build:

Convolution Layers, Activation, Pooling Layers, Fully Connected Layer.

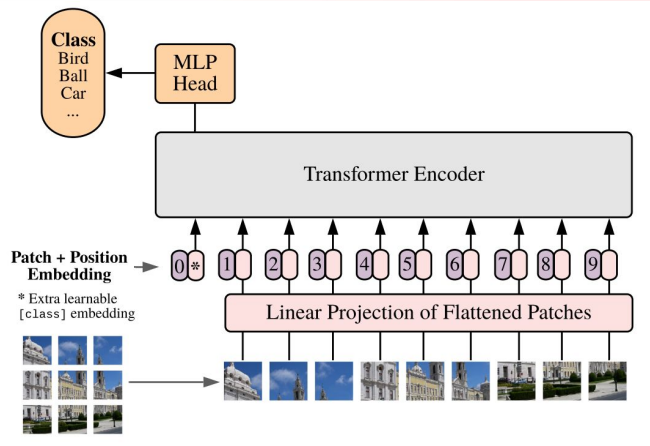
Properties:

Local Receptive Fields: Layers look at small, localized chunks of an image at a time rather than the whole thing at once.

Enforce strong inductive bias (translation invariance and locality)

Shared Weights: Instead of learning a new feature detector for every single corner of an image, a CNN learns one detector (like an "edge detector") and slides it across the entire image.

Spatial Hierarchy: The early layers detect simple things (edges, colors). The middle layers combine those edges into textures and motifs (eyes, wheels). The final layers piece those motifs into complex concepts (faces, cars).



Build:

Patch Embedding, Position Embedding, CLS token, Multi-Head Self Attention.

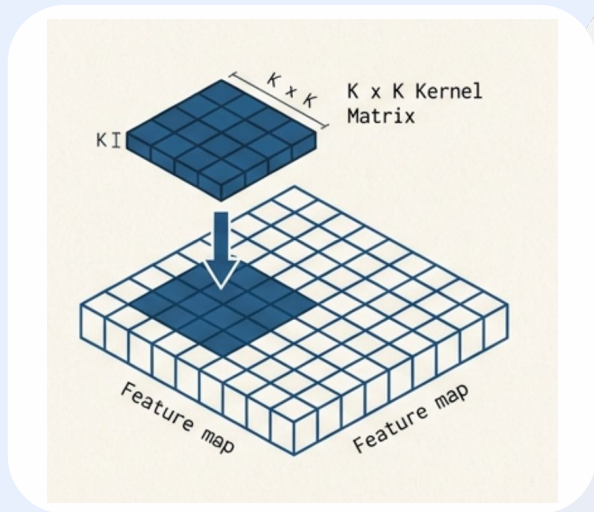
Properties:

A ViT views an image as a sequence of tokens, exactly like words in a sentence.

The model then uses Self-Attention to allow every token to look at every other token simultaneously.

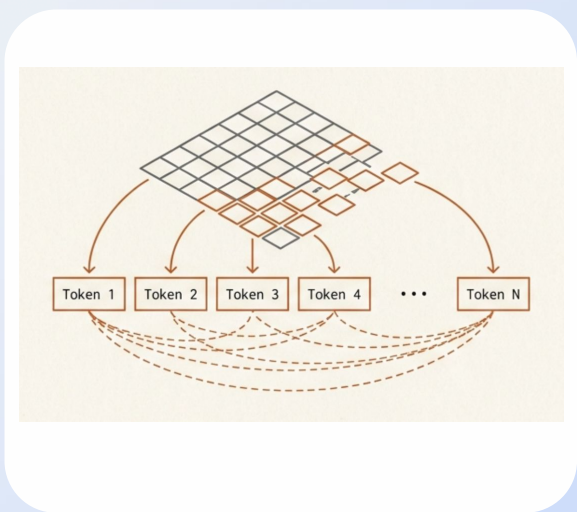
This allows the network to capture long-range dependencies and global context from the very first layer, something CNNs can only do after many downsampling steps.

The Core Paradigm Shift: Local Neighbourhood vs Global Sequences



CNNs:

- Receptive field: Localized, Grows with depth.
- Complexity Bounds: $O(H \times W \times C)$, Linear to resolution.
- Data dependency: Data efficient, works with small datasets.
- Scaling: Saturates early.
- Weakness: struggles with long range semantic dependencies.



ViTs:

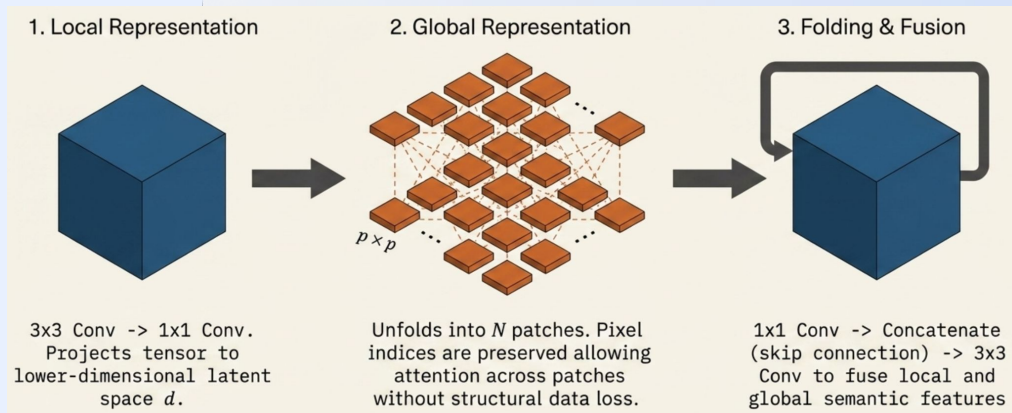
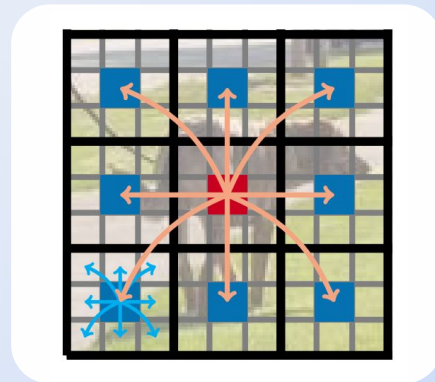
- Receptive field: Global from layer 1.
- Complexity Bounds: $O((H \times W)^2)$, Quadratic to token count.
- Data dependency: Data hungry.
- Scaling: Monotonic increase with compute.
- Weakness: Crushing memory/compute footprint for high resolution images.

Is it possible to combine the strengths of CNNs and ViTs to build a light-weight and low latency but highly capable network ?

MobileViT

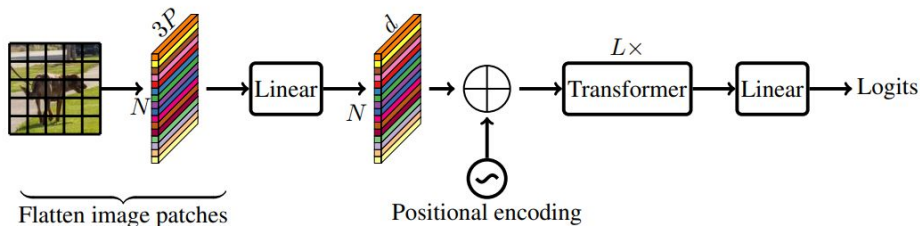
(Introduced by Apple AI Research)

- Light-weight, General Purpose, Low latency.
- Combines the benefits of CNNs (e.g., spatial inductive biases and less sensitivity to data augmentation) and ViTs (e.g., input-adaptive weighting and global processing).
- It treats the Transformer as a specialized layer within a standard convolutional network.
- MobileViT block replaces local processing in convolutions with global processing using transformers.

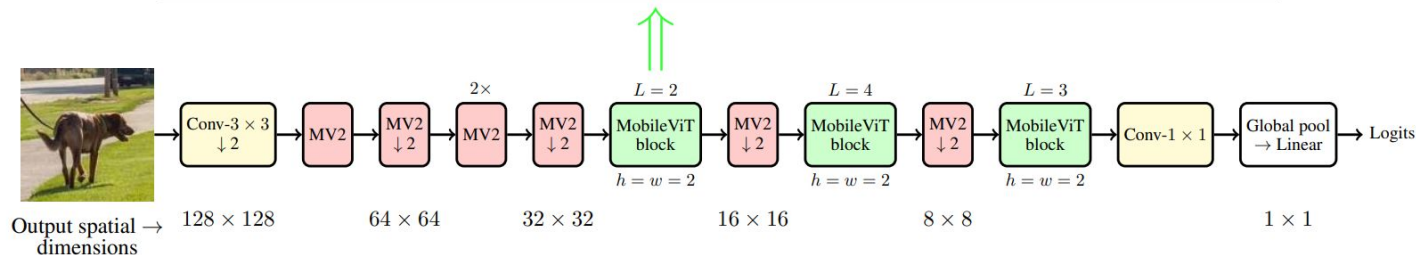
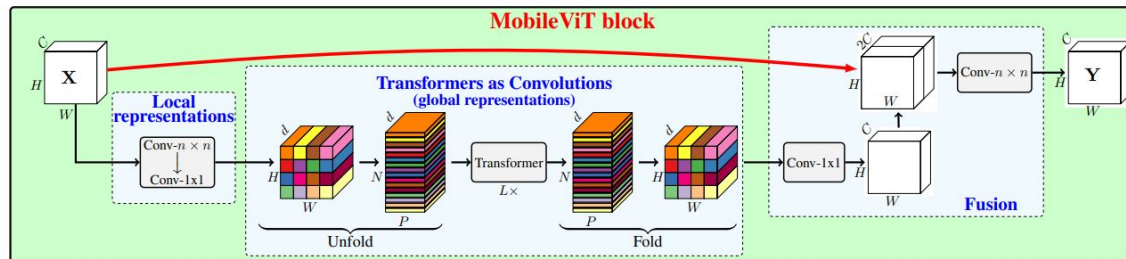


MobileViT: Architecture

Date



(a) Standard visual transformer (ViT)



(b) **MobileViT**. Here, $\text{Conv-}n \times n$ in the MobileViT block represents a standard $n \times n$ convolution and **MV2** refers to MobileNetv2 block. Blocks that perform down-sampling are marked with $\downarrow 2$.

Lets Trace the Tensor

Step 1: Local Convolution

input: $X \in \mathbb{R}^{H \times W \times C}$

$$X_L = \text{Conv}_{1 \times 1}(\text{Conv}_{n \times n}(X))$$

$$X_L \in \mathbb{R}^{H \times W \times d}$$

Step 2: Unfolding

$$X_U \in \mathbb{R}^{P \times N \times d}$$

$$N = \frac{HW}{P}$$

Patch size: $p \times p$

$$P = p^2$$

Step 3: Global Attention

$$X_G = \text{Transformer}(X_U)$$

$$X_G \in \mathbb{R}^{P \times N \times d}$$

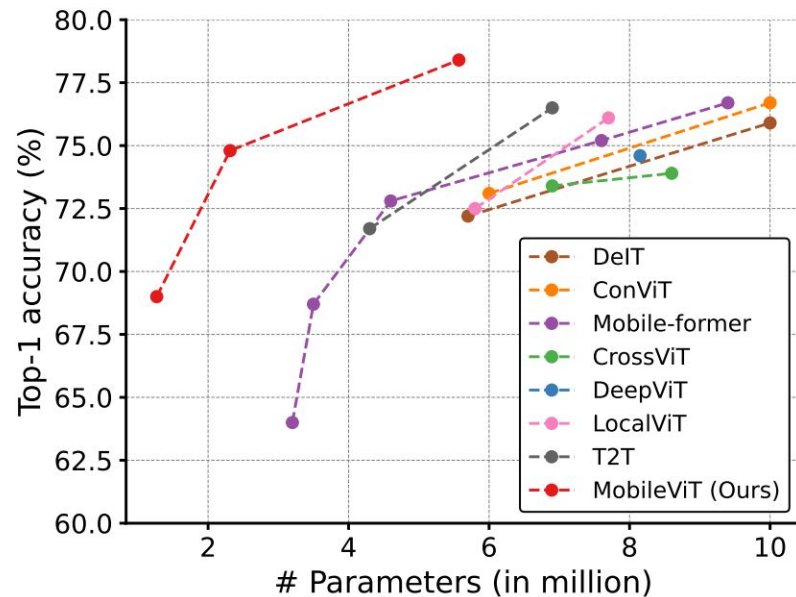
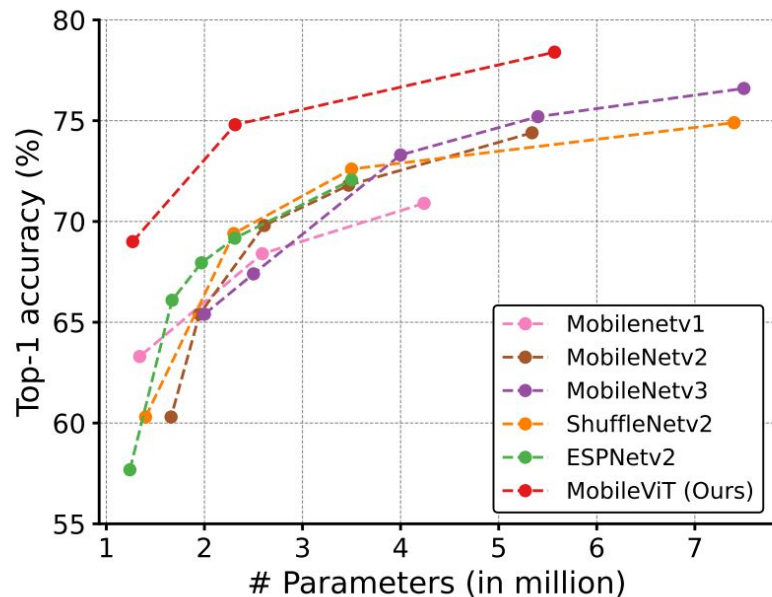
Step 3: Folding and Convolution Fusion

$$X_F = \text{Fold}(X_G) \quad \text{where} \quad X_F \in \mathbb{R}^{H \times W \times d}$$

$$Y = \text{Conv}_{n \times n}([X, \text{Conv}_{1 \times 1}(X_F)])$$

Comparison with CNNs and ViTs.

Date



Thank you

